AQ3.3: Activity Questions 3 - Not Graded

This assignment will not be graded and is only for practice.

Level 1

1 point

Which of the following vectors can be written as the linear combination of vectors $(2,3)$ and $(1,2)$ in R^2 with usual addition and scalar multiplication?

- [ ] $(1,0)$
- [ ] $(0,1)$
- [ ] $(0,0)$
- [ ] Any vector in R^2

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(1,0)$

$(0,1)$

$(0,0)$

Any vector in R^2

1 point

Suppose the set $\{(3,7,4), v\}$ is a linearly dependent set in the vector space R^3 with usual addition and scalar multiplication. Which of the following vectors are possible as a candidate for v?

- [ ] $(0,0,0)$
- [ ] $(5, \frac{35}{3}, \frac{20}{3})$
- [ ] $(1,0,0)$
- [ ] $(6,14,-8)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(0,0,0)$

$(5, \frac{35}{3}, \frac{20}{3})$

1 point

Suppose the set $\{(2,3,0), (0,1,0), v\}$ is a linearly dependent set in the vector space R^3 with usual addition and scalar multiplication. Which of the following vectors are possible as a candidate for v?

- [ ] $(2,3,1)$
- [ ] $(1,2,0)$
- [ ]

$(1, 3, 0)$(1, 3, 0)

☐

$(0, 1, 1)$(0, 1, 1)

☐

$(\pi, e, 0)$(π, e, 0)

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(1, 2, 0)$(1, 2, 0)

$(1, 3, 0)$(1, 3, 0)

$(\pi, e, 0)$(π, e, 0)

If $(3, 4) = \alpha(1, 2) + \beta(1, 3)$(3, 4) = α(1, 2) + β(1, 3) in the vector space R^2 R2 with usual addition and scalar multiplication, then find the value of $\alpha + 3\beta$.α + 3β.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) -1

1 point

If (9, 3, 1) is a linear combination of the vectors $(1, 1, 1), (1, -1, 1)$ and (x, y, z)(1, 1, 1), (1, −1, 1) and (x, y, z) as $2(1, 1, 1) + 3(1, -1, 1) + 4(x, y, z) = (9, 3, 1)$
2(1, 1, 1) + 3(1, −1, 1) + 4(x, y, z) = (9, 3, 1), then find the value of $x + y + 2z$x + y + 2z.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 0

1 point

1 point

Let VV be a vector space and v_1, v_2, v_3 and $v_4 \in V$v1, v2, v3 and v4 ∈ V . If v_1v1 is a linear combination of $v_i, i = 2, 3, 4$ $i.e.$ $v_1 = av_2 + bv_3 + cv_4$vi, i = 2, 3, 4 i.e. v1 = av2 + bv3 + cv4 then

☐

v_2v2 is a linear combination of $v_i, i = 1, 3, 4$vi, i = 1, 3, 4.

☐

v_3v3 is a linear combination of $v_i, i = 1, 4$vi, i = 1, 4.

☐

v_2v2 is a linear combination of v_2v2.

☐

v_4v4 is a linear combination of v_4v4.

No, the answer is incorrect.

Score: 0

Accepted Answers:

v_2v2 is a linear combination of v_2v2.

v_4v4 is a linear combination of v_4v4.

1 point

Consider three vectors $v_1 = (1, 0, 0), v_2 = (1, 1, 0), v_3 = (1, 1, 1)$v1 = (1, 0, 0), v2 = (1, 1, 0), v3 = (1, 1, 1) in the vector space R^3 R3, with usual addition and scalar multiplication. Which of the following sets is (are) true?

☐

$\{v_1, v_2, v_3\}$ is a linearly dependent set.

☐

$\{v_1 + v_2, v_1 + v_3, v_3\}$ is a linearly dependent set.

☐

$\{v_2 - v_1, v_1, v_2\}$ is a linearly dependent set.

☐

$\{v_1, v_2 - v_1, v_3 - v_2, v_3\}$ is a linearly dependent set.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{v_2 - v_1, v_1, v_2\}$ is a linearly dependent set.

$\{v_1, v_2 - v_1, v_3 - v_2, v_3\}$ is a linearly dependent set.

Level - 2

1 point

Let S be a subset of R^3 which is linearly dependent. Which of the following options are true?

☐

$S \cup \{(1, 0, 0)\}$ must be linearly dependent.

☐

$S \cup \{v\}$ must be linearly dependent for any $v \in R^3$.

☐

$S \setminus \{v\}$ must be linearly dependent for any $v \in R^3$.

☐

There may exist some $v \in S$, such that $S \setminus \{v\}$ is still linearly dependent.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$S \cup \{(1, 0, 0)\}$ must be linearly dependent.

$S \cup \{v\}$ must be linearly dependent for any $v \in R^3$.

There may exist some $v \in S$, such that $S \setminus \{v\}$ is still linearly dependent.

$M_{3\times 3}(R)$ denotes the vector space consisting of real square matrices of order 3 with usual matrix addition and scalar multiplication. Consider the following matrices:

$$M_1 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}, M_2 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}, M_3 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 1 & 2 & 1 \end{bmatrix}, M_4 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 1 & -1 & 1 \end{bmatrix}$$

$$M1 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}, M2 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}, M3 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 2 & -1 \end{bmatrix}, M4 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 1 & -1 & 1 \end{bmatrix}$$

Use the above information to answer questions 9 and 10.

1 point

Which of the following options are correct?

☐

$M_1 - M_2 = 0$ $M1 - M2 = 0$, where 0 denotes the zero matrix of order 3.

- [ ] $2M_1 + M_2 = M_3$ 2M1 + M2 = M3
- [ ] $2M_1 - M_2 - M_3 = 0$ 2M1 − M2 − M3 = 0 , where 00 denotes the zero matrix of order 3.
- [ ] $2M_2 - M_1 - M_4 = 0$ 2M2 − M1 − M4 = 0, where 00 denotes the zero matrix of order 3.
- [ ] The tuple (a, b, c) (a, b, c) is unique for which $aM_1 + bM_3 + cM_4 = 0$ aM1 + bM3 + cM4 = 0 holds, where 00 denotes the zero matrix of order 3.
- [ ] The pair (a, b) (a, b) is unique for which $aM_1 + bM_2 = 0$ aM1 + bM2 = 0 holds, where 00 denotes the zero matrix of order 3.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$2M_1 - M_2 - M_3 = 0$ 2M1 − M2 − M3 = 0 , where 00 denotes the zero matrix of order 3.

$2M_2 - M_1 - M_4 = 0$ 2M2 − M1 − M4 = 0, where 00 denotes the zero matrix of order 3.

The pair (a, b) (a, b) is unique for which $aM_1 + bM_2 = 0$ aM1 + bM2 = 0 holds, where 00 denotes the zero matrix of order 3.

1 point

Which of the following options is(are) true?

- [ ] $\{M_1, M_2\}$ {M1, M2} is a linearly dependent set.
- [ ] $\{M_1, M_2, M_3\}$ {M1, M2, M3} is a linearly dependent set.
- [ ] $\{M_1, M_2, M_4\}$ {M1, M2, M4} is a linearly dependent set.
- [ ] $\{M_1, M_3, M_4\}$ {M1, M3, M4} is a linearly dependent set.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{M_1, M_2, M_3\}$ {M1, M2, M3} is a linearly dependent set.

$\{M_1, M_2, M_4\}$ {M1, M2, M4} is a linearly dependent set.

$\{M_1, M_3, M_4\}$ {M1, M3, M4} is a linearly dependent set.

Check Answers

Your score is: 0/10